

Online Examination Report

Date:	17.07.2014
Attempts:	1
Examinee:	Okeng, Japheth
Date of birth:	
Subject:	040-Human performance
Result:	62,5 %
Time used:	00:30:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	174	40.	HUMAN PERFORMANCE	1,00	1,00
2	184	40.	HUMAN PERFORMANCE	0,00	1,00
3	205	40.	HUMAN PERFORMANCE	1,00	1,00
4	179	40.	HUMAN PERFORMANCE	0,00	1,00
5	170	40.	HUMAN PERFORMANCE	0,00	1,00
6	194	40.	HUMAN PERFORMANCE	0,00	1,00
7	20	40.1.2	Accident statistics	1,00	1,00
8	108	40.2	Basic aviation physiology and health maintenance	1,00	1,00
9	106	40.2	Basic aviation physiology and health maintenance	1,00	1,00
10	70	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
11	18	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
12	63	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
13	100	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
14	89	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
15	95	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
16	45	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
17	6	40.2.2.2	Vision	0,00	1,00
18	98	40.2.2.2	Vision	1,00	1,00
19	8	40.2.2.5	Integration of sensory inputs	1,00	1,00
20	30	40.2.3	Health and hygiene	0,00	1,00
21	107	40.2.3	Health and hygiene	1,00	1,00
22	32	40.2.3.3	Problem areas for pilots	1,00	1,00
23	4	40.2.3.3	Problem areas for pilots	1,00	1,00
24	72	40.2.3.4	Intoxication	1,00	1,00
25	87	40.2.3.4	Intoxication	0,00	1,00
26	121	40.3	BASIC AVIATION PSYCHOLOGY	1,00	1,00
27	124	40.3	BASIC AVIATION PSYCHOLOGY	1,00	1,00
28	9	40.3.1.3	Memory	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	65	40.3.1.4	Response selection	1,00	1,00
30	40	40.3.2.1	Reliability of human behaviour	0,00	1,00
31	21	40.3.2.4	Error generation	0,00	1,00
32	110	40.3.3	Decision making	1,00	1,00
33	56	40.3.4.3	Co-operation	1,00	1,00
34	5	40.3.4.3	Co-operation	0,00	1,00
35	57	40.3.4.4	Communication	0,00	1,00
36	127	40.3.5	Human behaviour	0,00	1,00
37	25	40.3.5.1	Personality, attitude and behaviour	0,00	1,00
38	7	40.3.6.1	Arousal	0,00	1,00
39	37	40.3.6.2	Stress	1,00	1,00
40	48	40.3.6.2	Stress	1,00	1,00
Total:62%				25,00	40,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Okeng, Japheth

Location: Uganda Civil Aviation Authority

Your Result: 25,00 from 40,00 Points (62,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
40. HUMAN PERFORMANCE	40	25,00	40,00	62,50
40.1. Human Factors: basic concepts	1	1,00	1,00	100,00
40.2. Basic aviation physiology and health maintenance	18	14,00	18,00	77,78
40.3. BASIC AVIATION PSYCHOLOGY	15	8,00	15,00	53,33
Total	40	25,00	40,00	62,50

1

One of the steps in the aeronautical decision making (ADM) process involved in good decision making is

- ☐ making a rational evaluation of the required actions.
- ☐ developing a 'can do' attitude.
- ☒ identifying personal attitudes hazardous to safe flight.

2

What is the antidote for a pilot with a 'macho' attitude?

☐ I'm not helpless. I can make a difference.

☐ Taking chances is foolish.

☒ Follow the rules. They are usually right.

3

Which procedure is recommended to prevent or overcome spatial disorientation?

- ☐ Rely on the kinesthetic sense.
- ☐ Reduce head and eye movements to the extent possible.
- ☒ Rely on the indications of the flight instruments.

4

Hazardous attitudes which contribute to poor pilot judgment can be effectively counteracted by

- ☒ early recognition of these hazardous attitudes.
- ☐ taking meaningful steps to be more assertive with attitudes.
- ☐ an appropriate antidote.

5

When should a flight instructor begin teaching aeronautical decision making (ADM) to a student?

- ☐ As soon as the student is able to control the aircraft during basic maneuvers.
- ☐ After the student has completed the initial solo flight but before conducting cross country flights.
- ☒ Beginning with the first lesson.

6

What visual illusion creates the same effect as a narrower-than-usual runway?

- ☐ An upsloping runway.
- ☒ A downsloping runway.
- ☐ A wider-than-usual runway.

7

The distribution of primary causes of accidents in the worldwide jet aircraft commercial fleet shows that human error is involved in:

- ☒ about 70% of cases
- ☐ 70% of cases
- ☐ all cases, one way or another
- ☐ about 90% of cases

8

Hypoxia susceptibility due to inhalation of carbon monoxide increases as

☐ oxygen demand increases.

☒ altitude increases.

☐ humidity decreases.

9

To overcome the symptoms of hyperventilation, a pilot should

- ☐ swallow or yawn.
- ☐ increase the breathing rate.
- ☒ slow the breathing rate.

10

Someone who has anaemia does not have

- ☐ enough plasma
- ☐ enough platelets
- ☒ enough functional haemoglobin
- ☐ enough white blood cells

11

The rate and depth of breathing is primarily controlled by:

- ☐ the amount of carbon monoxide in the blood
- ☐ the total atmospheric pressure
- ☒ the amount of carbon dioxide in the blood
- ☐ the amount of nitrogen in the blood

12

When flying above 10.000 feet hypoxia arises because:

- ☐ composition of the blood changes
- ☐ percentage of oxygen is lower than at sea level
- ☐ composition of the air is different from sea level
- ☒ partial oxygen pressure is below the critical value of 55mm/Hg

13

Positive G will cause the blood flow to the brain to:

- ☐ remain constant
- ☐ first increase, then decrease
- ☒ increase
- ☐ decrease

14

After SCUBA diving (more than 30 feet of depth) you have to wait a period of time before flying again. This period is at least:

☐ 12 hours

☐ 6 hours

☒ 24 hours

☐ 48 hours

15

**You climb from 0 to 50,000 ft and measure the decrease of the pressure per 5.000 ft.
The absolute difference in barometric pressure is greatest between:**

☐ 10.000 and 15.000 feet

☐ 5.000 and 10.000 feet

☒ 0 and 5.000 feet

☐ 45.000 and 50.000 feet

16

The total gas volume of the lung is the sum of:

1. tidal volume
2. inspiratory reserve volume
3. expiratory reserve volume
4. residual volume

Which of the following lists the correct combination?

☐ 1, 2 and 3

☒ 1, 2, 3 and 4

☐ 2 and 3

☐ 1 and 2

17

During poor weather conditions a pilot should fly with reference to instruments because:

- ☒ his attention will be distracted automatically under these conditions
- ☐ pressure differences can cause the altimeter to give wrong information
- ☐ perception of distance and speed is difficult in an environment of low contrast
- ☐ the danger of a "greying out" will make it impossible to determine the height above the terrain

18

The retina allows for colour perception as a result of the:

- ☐ rods located in its central part
- ☐ rods located in its peripheral zone
- ☐ crystalline lens
- ☒ cones located in its central part

19

Which sensations does a pilot get, when he is rolling out of a prolonged level turn?

- ☐ Turning into the original direction
- ☒ Turning in the opposite direction
- ☐ Climbing
- ☐ Flying straight and level

20

The following can be observed when the internal body temperature falls below 35°C:

- ☐ shivering, will tend to cease, and be followed by the onset of apathy
- ☒ mental disorders, and even coma
- ☐ profuse sweating
- ☐ the appearance of intense shivering

21

Which of these is true regarding the presence of alcohol within the human body?

- ☐ A small amount of alcohol increases vision acuity.
- ☒ Judgment and decision-making abilities can be adversely affected by even small amounts of alcohol.
- ☐ An increase in altitude decreases the adverse effect of alcohol.

22

Flying with a "common cold":

- ☐ is permitted as long as you are on treatment with antibiotics.
- ☐ increases the risk of hypertension.
- ☒ may lead to incapacitation due to severe sinus or ear pain.
- ☐ will cause infection in other crew members if you are flying in a pressurised aircraft.

23

How is Yellow fever contracted?

- ☐ Excessive use of alcohol.
- ☐ By contact with the saliva of infected animals.
- ☐ Contaminated food or water.
- ☒ A virus transmitted by an infected mosquito.

24

Alcohol degrades:

☐ Stage 3 and 4 sleep

☒ Paradoxical sleep

☐ Stage 2 sleep

☐ Slow wave sleep

25

Which statement is correct?

1. Smokers have a greater chance of suffering from coronary heart disease
2. Smoking tobacco will raise the individuals physiological altitude during flight
3. Smokers have a greater chance of contracting lung cancer

☐ 1 and 2 are correct, 3 is false

☐ 2 and 3 are correct, 1 is false

☒ 1 and 3 are correct, 2 is false

☐ 1,2 and 3 are correct

26

To help manage cockpit stress, pilots must

- ☐ be aware of life stress situations that are similar to those in flying.
- ☐ avoid situations that will improve their abilities to handle cockpit responsibilities.
- ☒ condition themselves to relax and think rationally when stress appears.

27

While conducting an operational check of the cabin pressurization system, the pilot discovers that the rate control feature is inoperative. He knows that he can manually control the cabin pressure, so he elects to disregard the discrepancy. Which of the following alternatives best illustrates the INVULNERABILITY reaction?

- ☐ It's too late to fix it now.
- ☐ He can handle a little problem like this.
- ☒ What is the worst that could happen.

28

Information stays in the short-term memory

- ☐ less than 1 second
- ☐ around 24 hours
- ☐ from 5 to 10 minutes
- ☒ about 20 seconds

29

The ability to monitor information which could indicate the development of a critical situation

- ☐ is dangerous, because it distracts attention from flying the aircraft
- ☐ is responsible for the development of inadequate mental models of the real world
- ☐ makes no sense because the human information processing system is limited anyway
- ☒ is necessary to maintain good situational awareness

30

Situations particularly vulnerable to "reversion to an earlier behaviour pattern" are:

1. when concentration on a particular task is relaxed
2. when situations are characterised by medium workload
3. when situations are characterised by stress

☐ 1. and 3.

☒ 3 only.

☐ 2. and 3.

☐ 1. and 2.

31

Human error rates during the performance of a simple and repetitive task can normally be expected to be approximately:

☐ 1 in 200

☐ 1 in 2000

☒ 1 in 500

☐ 1 in 100

32

Define Aeronautical Decision Making(ADM)

- ☐ decision making process which relies on good judgment to reduce risks associated with each flight.
- ☐ mental process of analyzing all information in a particular situation and making a timely decision on what action to take.
- ☒ systematic approach to the mental process used by pilots to consistently determine the best course of action for a given set of circumstances.

33

Expressed as a simple mathematical statement when considering a crew of two, good synergy is:

☐ $1 + 1 = < 2$

☐ $1 - 1 = > 2$

☒ $1 + 1 = > 2$

☐ $1 +/- 1 = < 2$

34

Which of the following statements best characterises a self-centered cockpit ?

- ☒ While decreasing communication, the independence of each member bolsters the crew's synergy
- ☐ Each one does his own thing while at the same time assuming that everyone is aware of what is being done or what is going on
- ☐ The communication between crew members always increases when the captain takes charge of a situation
- ☐ The egoistic and self-centered personality of the captain often leads to a synergetic cockpit

35

Which statement is correct?

- ☐ Inconsistent communication improves flight safety.
- ☐ There is no relation between inadequate communication and incidents or accidents.
- ☐ Problems in the personal relationships between crew members are likely to hamper the communication process.
- ☒ Personal conflict that takes place prior to take-off should wait to be addressed until the end of the flight.

36

When a pilot recognizes a hazardous thought, he or she then should correct it by applying the corresponding antidote. Which of the following is the antidote for ANTIAUTHORITY/DON'T TELL ME hazardous attitude?

☐ Don't tell me. Follow the rules. They are usually right.

☐ It won't happen to me. It could happen to me.

☒ Not so fast. Think first.

37

Contrary to a person's personality, attitudes:

- ☐ are non-evolutive adaptation procedures regardless of the result of the actions associated with them
- ☒ form part of personality and, as a result, cannot be changed in an adult
- ☐ Are the product of personal disposition and past experience with reference to an object or a situation
- ☐ are essentially driving forces behind changes in personality

38

An identical situation can be experienced by one pilot as exciting in a positive sense and by another pilot as threatening. In both cases:

- ☐ both pilots will loose their motor-coordination
- ☐ the pilot feeling threatened, will be much more relaxed, than the pilot looking forward to what may happen
- ☒ both pilots will experience the same amount of stress
- ☐ the arousal level of both pilots will be raised

39

Anxiety can affect:

- 1. Judgement**
- 2. Attention**
- 3. Memory**
- 4. Concentration**

☐ 1 only is correct

☐ 1 and 2 only are correct

☒ All are correct

☐ 1, 2 and 4 only are correct

40

The behavioural effects of stress may include:

- 1: manifestation of aggressiveness.**
- 2: a willingness to improve communication.**
- 3: a willingness for group cohesion.**
- 4: a tendency to withdrawal.**

The combination of correct statements is:

- ☐ 2,3 and 4 are correct
- ☒ 1 and 4 are correct
- ☐ 1,2 and 3 are correct
- ☐ 3 and 4 are correct